Learning by competing: Competitive Multi-generators based Adversarial Learning

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Abstract—Generative adversarial networks (GANs) have been extensively used for dozens of image enhancement and image translation applications where several traditional and novel architectures have been and are still being introduced. However, the classical training protocols do not fully explore the significant potential of such architectures. In this paper, we propose a novel GAN framework called competitive multi-generator based GAN (CMGAN) for more effective training progress and enhanced image generation. Different from classical adversarial learning protocols, the generators of a CMGAN are not only challenged by the discriminator for improved performance, but by the other generators also. This is achieved by a sharing mechanism, where the generators' current performances are shared among each other, motivating each generator to perform better than the rest. Extensive experiments in several conditional image generation applications show the significant improvements achieved by several traditional architectures when trained following our proposed learning strategy. Moreover, these architectures, trained by our learning strategy, are able to outperform the state-of-the-art approaches proposed in each tested application.

Index Terms—GANs, multi-generator, competition, supervised image generation.

1 Introduction

GANs are one of the most promising architectures in the field of deep learning. GANs are designed as a minimax game between two networks [1]. The first network, called the generator, is trained to fool the second network, called the discriminator, by generating instances that should have the same statistical distribution as the real ones. Alternately, the discriminator is trained to distinguish between real instances and fake ones generated by the generator. The game ends when the discriminator fails to distinguish between real and fake instances (Nash Equilibrium). Due to their efficiency, GANs are widely implemented in almost every application introduced to generate images based on predefined objectives [2], [3], [4], [5]. Based on the target goal, GANs generate images following two main approaches: 1) unsupervised image generation and 2) supervised (conditional) image generation. In the first approach, GANs transform images from one domain (source) to another domain (target), using datasets with unpaired images for training such GANs [6], [7], [8]. Following the second approach, GANs generate images from the same domain as the target by conditioning the networks on additional information that directs the generation process towards certain target distributions (groundthruth), using datasets with paired images for training [4], [9], [10].

Regardless of the training approach followed, different GANs architectures are proposed so that the player base of the minimax game is expanded, involving multiple generators and discriminators instead of a single pair of them [11], [12], [13], [14]. Based on this fact, the majority of the proposed GANs variants are trained following three main approaches: 1) multiple generators vs. single discriminator [12], [14], 2) single generator vs. multiple discriminators [15], [16], and 3) multiple generators vs. multiple discriminators [17], [18]. In this paper, the discussion is limited to the cases where multiple generators are used. Several studies [11], [19] support the fact that the use of multiple generators can significantly reduce the minimax gap, with greater stability observed when training such GANs. Moreover, the inclusion of multiple generators in an unsupervised image generation task is an essential stage to avoid the mode collapse challenge, where a GAN with a single generator tends to generate images belonging to a specific mode rather than all modes. Therefore, GANs with multiple generators were introduced to solve this problem, and dozens of GAN architectures, as well as objective functions, were proposed to guarantee a more diverse image generation [11], [13]. However, incorporating multiple generators for unsupervised image generation does not guarantee avoiding the mode collapse problem. Therefore, numerous strategies have been presented to force the generators to generate images that are fundamentally different from the images produced by other generators [20], [21].

When it comes to the task of conditional image generation, GANs with multiple generators cannot be trained using the same objectives or protocols as used to train unsupervised GANs. While the generators of unsupervised GANs are motivated to generate images with different modes, the generators of conditional GANs should generate images that are similar to the corresponding given ground-truths (condition). In other words, the generators of unsupervised GANs are designed to collectively play the minimax game against the discriminator while generating images whose modes differ from each other. On the other hand, the generators of conditional GANs are usually chained to perform a multilevel enhancement of input images, where chained generators are expected to perform better than a single generator.

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In light of all these concerns, in this paper, we propose a conditional GANs based framework in which the generators play two minimax games instead of one. In the first minimax game, the participating generators compete against the discriminator, where the generators jointly try to fool the discriminator. The second minimax game involves the generators themselves, where each generator tries to outperform other generators. The proposed competitive objective of such GANs is to motivate all participating generators to perform at their best in terms of accuracy and convergence speed. Although our proposed approach is applicable to both conditional and unsupervised image generation, the main focus is on conditional image generation/translation. To explore the solutions required to maximize the generation performance of a multi-generator GAN, different motivational strategies and architectures are investigated. In summary, the contributions of the paper can be described as follows.

1) We propose a competitive training framework, so the participating models are trained in a competitive fashion to outperform themselves as well as other models.
2) We design a unique loss function that encourages each participating model to outperform the other competitors.
3) In addition to the classical baseline GAN architectures, we introduce a new auto-encoder architecture that takes advantage of U-Net and the inception module.
4) Extensive experiments that examine different competitive training protocols (*one-vs-one* and *one-vs-all*) in which participating models are evaluated and tested in various conditional image generation applications such as desnowing, dewatering, deblurring, and others.

2 RELATED WORK

Enhancing the performance of GANs by incorporating multiple generators into the generation process has attracted great attention since the first GAN was proposed. Ghosh et al. [14] proposed a multi-agent unsupervised GAN. Their generators are driven by a message-sharing strategy in which the generators are motivated to outperform themselves while generating images with different modes. Moreover, the generators are guided by two different objectives: the competing objective and the conceding objective. The first objective motivates each generator to fool the discriminator more than other generators. On the contrary, the second objective motivates the generators to fool the discriminator in a cooperative manner. Hoang et al. [12] presented a multi-generator GAN with a single discriminator and a single classifier that helps the generators produce more diverse image modes by distinguishing the specific generator of a given image. The objective of the model guides the pool of generators to collectively generate images with the same distribution of the target dataset. In addition, there is a softmax loss that takes advantage of the classifier to encourage generators to produce images that are different from those of other generators. Later, Ghosh et al, [14] proposed a multi-agent GAN that integrates classification functionality into the discriminator, whose output is modified to produce softmax scores representing the probability that an image was generated by a given generator. Chavdarova and Fleuret [13] presented a multi-generator and multi-discriminator GAN, which is divided into an ensemble of local GANs pairs and a global GANs pair. The local GAN pairs are trained independently following the classical approach, while the global generator is trained to fool all local discriminators. Additionally, the global discriminator is trained with images generated by all local generators. Such a strategy helps the proposed approach avoid oscillations during training, in addition to the mode collapse problem. Seeking more training stability and better generation, Wang et al. [22] introduced the first GAN that fully exploits the evolutionary learning mechanism. Their proposed GAN consists of a set of generators known as population, which try to adapt to the environment represented by the discriminator. To achieve better adaptation, multiple variations (children) are extracted from each generator by considering different loss objectives such as minimax loss, heuristic loss, and least squares loss. As in any evolutionary narrative, the variations that manage to fool the discriminator more and produce more diverse images are kept for another evolutionary cycle, while the weak variations are discarded.

As mentioned earlier, multi-generator GANs can also be designed so their generators are chained as a multi-stage enhancement approach to reach a predefined target. Quan et al. [23] introduced an additional refinement stage in their proposed approach to MRI image reconstruction, where another generator is added to accept the output of the first generator. Both generators are trained to reach the same groundtruth so that two generated images are produced from a single input image. Tang et al. [24] proposed a GAN with chained generators consisting of an encoder with common parameters and two independent decoders (generators); the first generates edge maps and the second generates RGB images. Edge map features are integrated into the images generated by the first generator to enhance the generation of more realistic images. A multimodality discriminator is employed to distinguish the real and fake images at both edge and pixel levels. Wen et al. [25] presented a chained-generators GAN that contains two generators and two discriminators, where the output of the first generator is fed into the second generator. The first generator is trained to produce images with good motion prediction, whereas the second generator is trained to produce images with better fine details. Special loss functions are defined to guide each generator in generating its best results. Recently, Phan et al. [10] presented a chained-generators GAN that performs multi-stage refinement of noisy input speech signals. To achieve this purpose, two parameter sharing scenarios were tested. In the first scenario, the chained generators share their parameters during training, while in the second scenario, where no parameter sharing occurs, two mappings are learned independently. The loss function is designed in a way so a given generator is learned to generate a sample that is twice as good as the previous one.

3 PROPOSED METHOD

The main goal of our proposed approach is to provide a competitive environment that motivates the generators of a competitive multi-generator GAN (MGAN) to compete

with each other. Thus, each generator of CMGAN should be aware of the performance of other generators to attempt to outperform them. Below, we, respectively, introduce the concept of a competitive MGAN, the participating competitors, and the competition criteria defined so the competitors can be trained in a competitive fashion.

3.1 Generative adversarial networks

The generative adversarial networks have been proposed for the first time by Goodefellow [1] as a tool to generate new images that look like images that belong to the target distribution and have its distribution. To achieve this purpose, a generator network $\mathbf{G}$ is employed to regenerate samples that share the same data distribution of the target data. Additionally, a discriminator network $\mathbf{D}$ is incorporated to measure the probability of the generated samples having the same data distribution or not. Based on a game theoretic min-max principle, the generator and discriminator are jointly learned by alternating the training of $\mathbf{D}$ and $\mathbf{G}$.

3.2 Overview of the proposed CMGAN

In the majority of the conditional image generation applications, traditional multi-generator GANs are designed so a minimax game is played between a set of generators and a discriminator where the generators are motivated to work collaboratively to generate good results. Inspired by the fact that emphasizes the role of competitiveness among human players as an important game mechanics, we propose to let the generators of a multi-generator GAN mimic such competitive behavior. The core idea is simple yet efficient, the minimax game that is usually played between a set of generators and a discriminator is now extended so the participating generators are involved in sub-minimax games among each others. Each generator is motivated not only to generate fake images close to the real ones, but also to generate images better than other generators. Competitiveness is expected to contribute positively to the performance of each generator, and the results estimated by a generator that is trained following a competitive training protocol are expected to be better than the results estimated by the same generator if it is trained independently.

3.3 competitors

Like any competitive game, competitiveness becomes more effective when players have different characteristics and qualifications. Therefore, the participating generators should have different network architectures and different generation abilities. Two of the widely used generator network architectures are the U-Net [26] and ResNet-Blocks [27] autoencoders. The concept of U-Net architecture is to supplement downsampling stream by symmetric layers (decoding) where traditional pooling operations are replaced by transposed convolution based upsampling giving the final architecture the shape of letter U. As a result, these layers increase the resolution of the decoded output with more precision thanks to the skip connections. On the other hand, ResNet-based generator networks adopt residual blocks to compute relevant features from the input images. Hence, the input image is downsampled multiple times before the resulted convoluted feature maps are fed into residual subnetworks referred to as ResNet-Blocks. Afterwards, the output of the ResNet-Blocks is upsampled to reach the desired size of the generator output.

In addition to these classical competitors, we introduce a third competitor, called a U-shaped U-IncNet-based network (U-IncNet) that includes additional feature extraction blocks which link the downsampling and upsampling streams. The Sub-blocks for feature extraction involve several Inception.v3 [28] modules. The introduced U-IncNet.v3 module computes the convolution responses with multi-kernel sizes: $(1 \times 1), (3 \times 3), (5 \times 5), (7 \times 7)$. These filters are concatenated to form the final output of the Inception.v3 module. The choice of Inception.v3 module as a feature extractor is related to its ability to capture visual features at different scales thanks to its different kernel sizes. Figure **??** shows the architectures of the three considered baseline competitors: U-Net, ResNet, and the proposed U-IncNet.

3.4 competitive generators

According to [29], competitiveness is mainly guided by two main motivation factors: 1) the motivation for improving performance (MIP), and 2) the motivation to win (WIN). Motivated by the MIP factor, each competitor tries to outperform itself by enhancing its performance over time, while the WIN factor motivates each competitor to establish superiority over other competitors by outperforming their performances. To mimic these factors into our proposed CMGAN, we carefully designed the objective function taking these factors into consideration. But, we have to understand first the representation of the components of a traditional GAN's objective function. Classically, GANs are trained to generate an output image y mapped through G from a random noise vector z,where $G: \ z \rightarrow y$ and the objective function of such a GAN is given as follows

$$\min_{G} \max_{D} \mathcal{L}_{GAN}(G, D) = \mathbb{E}_{x,y}[log\ D(x, y)] + \mathbb{E}_{x,z}[log(1 - D(x, G(z)))] \quad (1)$$

where x represent the real data images that help training the discriminator distinguishing between real and fake images. In conditional image generation applications, conditional GANs [30] are usually used to generate an output image y mapped from both an input image x and a random noise vector z where $G: \ \{x, z\} \rightarrow y$, and the objective function of conditional GANs is given as follows

$$\min_{G} \max_{D} \mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x,y}[log\ D(x, y)] + \mathbb{E}_{x,z}[log(1 - D(x, G(x, z)))] \quad (2)$$

The objective function is designed so the generator G tries to minimize its objective function penalty via generating more realistic images that may fool the discriminator D whose objective is to distinguish fake images from real ones. In other words, the generator G and the discriminator D play the minimax game competitively. For better conditional image generation, numerous approaches and experiments [30], [31] have proven that the use of classical losses such as $\mathcal{L}_1$ or $\mathcal{L}_2$ in addition to the adversarial loss $\mathcal{L}_{cGAN}$; will

significantly enhance image generation. Therefore, the final objective function of a conditional GAN can be expressed as follows

$$\acute{G} = \min_{G} \max_{D} \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_1 \quad (3)$$

where in particular, the objective of the generator G is to minimize the following function

$$\mathcal{L}_G = \mathbb{E}_{x,z}[log(1 - D(x, G(x, z)))] + \lambda \mathcal{L}_1 \quad (4)$$

Seeking for simplicity, Eq. (4) is rewritten as follows

$$\mathcal{L}_G = \mathcal{L}_{adv} + \lambda \mathcal{L}_1 \quad (5)$$

Thus, the generator G is motivated not only to compete against the discriminator D, but also to improve its performance by generating images as close as possible to their corresponding groundtruths from the perspective of $\mathcal{L}_1$. Hence, loss $\mathcal{L}_1$ simply represents the MIP factor that is required to design GANs with competitive generators where each generator is motivated to produce better images regardless the results of other generators. On the other hand, the WIN factor which represents the motivation to win, is not yet defined or introduced in the classical objective function. To achieve this, a WIN loss that depends on the difference between the generators' individual performance is introduced in the objectives of all competing generators. This difference can be simply measured based on the $\mathcal{L}^i_{adv}$ loss and $\mathcal{L}_1$ loss. Subsequently, the generator with the lowest loss values in terms of $\mathcal{L}^i_{adv}$ and $\mathcal{L}_1$ can be considered as the generator with the best generation performance. However, a superiority of a given generator with respect to a certain loss does not guarantee a superiority with respect to the other loss. Hence, to motivate all participating generators to outperform each other, generators with poor generation performance with respect to a certain loss are penalized with the difference/s between their performance and the performance of the generator/s that outperforms respect to the same loss. seeking clarity, the WIN losses $\mathcal{L}^1_{WIN}$ and $\mathcal{L}^2_{WIN}$ that are incorporated in the objective functions of two competing generators G_1 and G_2 are respectively defined as follows:

$$\mathcal{L}^1_{WIN} = max(\nabla_{\mathcal{L}^1_{adv}, \mathcal{L}^2_{adv}}, 0) + \lambda \, max(\nabla_{\mathcal{L}^1_1, \mathcal{L}^2_1}, 0) \quad (6)$$

$$\mathcal{L}^2_{WIN} = max(\nabla_{\mathcal{L}^2_{adv}, \mathcal{L}^1_{adv}}, 0) + \lambda \, max(\nabla_{\mathcal{L}^2_1, \mathcal{L}^1_1}, 0) \quad (7)$$

where $\nabla_{\mathcal{L}^1_1, \mathcal{L}^2_1} = \mathcal{L}^1_1 - \mathcal{L}^2_1$ and $\nabla_{\mathcal{L}^1_{adv}, \mathcal{L}^2_{adv}} = \mathcal{L}^1_{adv} - \mathcal{L}^2_{adv}$ represents the difference between the performances of both competing generators in terms of $\mathcal{L}_{adv}$ and $\mathcal{L}_1$ respectively. The max operator guarantees penalizing a given generator by $\nabla_{\mathcal{L}^1_{adv}, \mathcal{L}^2_{adv}}$ and/or $\nabla_{\mathcal{L}^1_1, \mathcal{L}^2_1}$ when it is outperformed by the other competitor, and awarding the same generator by zero penalty when it outperforms the same competitor. Thus, this scenario can be generalized to a scenario of n ($n > 2$) generators, and the loss $\mathcal{L}^i_{WIN}$ that depends on the differences between the individual $\mathcal{L}_1$ losses of the generators n is introduced in the objectives of all competing generators as follows

$$\mathcal{L}^i_{WIN} = \sum_{j \neq i}^{n} max(\nabla_{\mathcal{L}^i_{adv}, \mathcal{L}^j_{adv}}, 0) + \lambda \sum_{j \neq i}^{n} max(\nabla_{\mathcal{L}^i_1, \mathcal{L}^j_1}, 0) \quad (8)$$

and the final loss function that guides each generator is defined as follows

$$\mathcal{L}_{G_i} = \mathcal{L}^i_{adv} + \lambda \mathcal{L}^i_1 + \mathcal{L}^i_{WIN} \quad (9)$$

Where $\nabla_{\mathcal{L}^i_{adv}, \mathcal{L}^j_{adv}} = \mathcal{L}^i_{adv} - \mathcal{L}^j_{adv}$ represents the difference between the adversarial loss of generator G_i and the adversarial loss of each generator G_j individually, while $\nabla_{\mathcal{L}^i_1, \mathcal{L}^j_1} = \mathcal{L}^i_1 - \mathcal{L}^j_1$ represents the difference between the $\mathcal{L}_1$ loss of generator G_i and the $\mathcal{L}_1$ loss of each generator G_j individually. The loss $\mathcal{L}^i_{WIN}$ represents the WIN factor that guarantees that each generator G_i is penalized by the difference values $\nabla_{\mathcal{L}^i_{adv}, \mathcal{L}^j_{adv}}$ and $\nabla_{\mathcal{L}^i_1, \mathcal{L}^j_1}$ if other generators perform better than it in terms of adversarial loss and $\mathcal{L}_1$ loss, while it returns zero penalty when the generator G_i outperforms all other generators.

3.5 Referee

As mentioned above, the discriminator task is to effectively distinguish the real images from the fake ones generated by the participating generators. This is achieved by Eq. (2) where the discriminator aims to maximize the probability score $D(x, y)$ produced when recognizing real images and minimize the probability score $D(G(x, z))$ produced when recognizing fake images. In the case of multi-generator GANs, Eq. (2) is extended so the discriminator judges the quality of the generated images by n generators

$$\max_{D} \mathcal{L}_{cGAN}(G_i, D) = \mathbb{E}_{x,y}[log \; D(x, y)] + \sum_{i=1}^{n} \mathbb{E}_{x,z}[log(1 - D(G_i(x, z)))] \quad (10)$$

However, in the case of conditional GANs, the impact of the discriminator in helping the generators to generate better images is downgraded by giving more weight to $\mathcal{L}_1$ loss compared to low values of adversarial loss $\mathcal{L}_{adv}$ that range between 0 and 1. Taking this fact into consideration, we can picture the discriminator of a competitive multi-generator GAN as a referee who ensures smooth performance of all generators and guarantees that their generations go in the right direction. In this paper, the discriminator network D relies on a pixel classification architecture. It is based on a set of convolutions and activation layers that lead to a binary label (fake/real) for each pixel. The state-of-the-art discriminator networks demonstrated good performances in detecting fake and real images, and then granting prominent adversarial loss to the generator. In our work, we use the Pixel Discriminator architecture proposed in [30].

3.6 Training strategies

In a competitive environment, the number of the participating competitors is expected to affect their performance. Therefore, a CMGAN with three generators -for example-

can be trained following two learning protocols: 1) *one-vs-one* and 2) *one-vs-all*. Following the first scenario, the generators are trained competitively in pairs where each generator is trained twice competitively, each time against only one of the other two generators. On the other hand, following the second scenario, all selected generators are trained jointly in a competitive fashion where each generator is motivated to outperform the other two generators simultaneously. Figure **??** illustrates the architectures of both competitive adversarial learning protocols while more interesting findings are revealed in the experiment section.

4 Experiments

To explore the effectiveness of the proposed CMGAN framework, we tested it in a variety of image translation and image enhancement applications. To measure the improvement achieved by the proposed competitive adversarial learning, a comprehensive ablation study that concerns the performance of the participating generators when trained following both individual and competitive learning protocols; is performed. Hence, the three selected competitors are first trained individually. Subsequently, each competitor is trained following the training strategies mentioned above, i.e. *one-vs-one* and *one-vs-all* learning scenarios. Also, the results of the best generator trained following these protocols are compared with the results estimated by several state-of-the-art techniques in each application. It is worth mentioning that regardless of the target application, the competing generators involved in these experiments are implemented as originally proposed in the literature. In other words and in contrast to the state-of-the-art approaches, neither the architectures nor the loss functions of such competing generators are uniquely designed to tackle the different challenges introduced in each application. For simplicity, the competing generators will be referred to in the discussion using the corresponding names of the employed auto-encoder architectures. For example, a U-IncNet based generator is referred to as U-IncNet.

4.1 Targeted applications (Ablation Study)

Due to the challenges they introduce, the proposed CMGANs are tested on the following applications :

- Desnowing: snowy image → snow-free image
- Deraindroping: raindrops image → raindrops-free image
- Infrared translation: RGB image → infrared image
- Deblurring: motion-blurred image → sharp image
- Dewatering: water-hazed image → clean image
- Demoireing: moiré image → clean image

The aforementioned applications are described in details in the following subsections. Except the experiment of dewatering where non-reference evaluation metrics are used, in all remaining experiments, the state-of-the-art deep learning approaches including the proposed CMGANs are evaluated using the standard Peak Signal-to-Noise Ratio (PSNR) and, additionally, the Structural Similarity (SSIM).

4.1.1 Desnowing

The objective of a desnowing approach is to successfully recover the original image from a snowy one by removing the snowflakes that come in different sizes, shapes, and degrees of transparency. Classically, such task was handled by hand-crafted techniques before being effectively dealt by deep learning approaches. The benchmark dataset that is used by most deep learning approaches is Snow100k dataset **??**. In this dataset snowy images were synthesized using 5.8k snow masks. Both training and testing sets are divided based on the size of the snowflakes into three subsets: Snow 100k-S (small snowflakes), Snow 100k-M (small and medium snowflakes), Snow 100k-L (small, medium, and large snowflakes).

Table **??** lists the average PSNR and SSIM metrics on Snow 100k-S dataset computed over the three test subsets. The maximum performance (PSNR/SSIM) achieved by an individual learning configuration is 30.35/0.9371 by U-IncNet, followed by U-Net with scores of 29.98/0.9372, and ResNet with scores of 28.83/0.9312. The proposed competitive adversarial learning leads to performance improvements that range from 1.21% to 2.66% and from 0.25% to 0.88% in terms of PSNR and SSIM, respectively. The highest performance reported is 31.53/0.9454 by U-IncNet when competing against U-Net. Moreover, U-IncNet appeared to be more open for improvement through competition, as it obtains the highest performance gains (3.88%) in contrast to ResNet whose performance is slightly improved (1.73%) and U-IncNet whose highest gain (2.66%) is achieved against ResNet.

4.1.2 RainDrop

When raindrops become adhered to a glass window, windscreen, or camera lens, they obstruct the view and degrade the image's clarity. Given the structure of raindrops which is similar to that of a fish-eye lens, raindrop regions are generated by rays of reflected light from a surrounding scope. Also, the camera's focus is often on the background area, blurring the look of raindrops. Raindrop images degrade visibility, so researchers have proposed several deep learning approaches based on various GANs versions to alleviate this problem. We selected the well-known Raindrop dataset [32], which consists of 1119 pairs of images captured with various background scenes and raindrops. In our experiment, the PSNR and SSIM metrics are the key performance indicators for evaluating the proposed CMGAN and comparing it with state-of-the-art approaches. We followed the same evaluation protocol described in [32].

Table **??** shows the PSNR and SSIM rates of the baseline competitors on RainDrop dataset. The ResNet-vs-All-Learning setup achieves the best performance of 31.85/0.9562 with an improvement of 1.43%/0.18% over the results of ResNet when trained individually. Competing against U-Net and U-IncNet generators individually, ResNet improves its PSNR/SSIM performance by 0.86%/0.17% and 0.67% /0.15%, respectively. Following both competitive learning protocols, U-Net achieves comparable performance to U-Net's standalone results, where the best slight improvements (0.07% /0.03%) over PSNR and SSIM scores are achieved by U-Net (vs. U-IncNet) architecture. Also, U-IncNet slightly improves its performance following *one-vs-all* learning protocols with a gain of 0.84%/0.65% in terms of PSNR/SSIM respectively.

4.1.3 Image Demoireing

The task of this application is to restore clean versions of images that are degraded by several patterns of moiré patterns, which are mainly the results of the interference between the camera's pixel grid and the electrical device screen. Several deep learning approaches have been proposed for moiré pattern removal where they were trained and tested on several datasets. The benchmark dataset that is used in our experiment is TIP2018 [33]. This dataset contains 135,000 image pairs where each pair consists of an image degraded by a moiré pattern and its corresponding non-degraded reference image. TIP2018 dataset offers a wide variety of moiré effects where 90% of the image pairs are used for training and validation while 10% of them are used for testing. Both PSNR and SSIM metrics are employed to evaluate the state-of-the-art approaches in addition to the proposed GANs.

Table **??** reports the PSNR and SSIM scores achieved by the baseline competitors on TIP2018 dataset following the aforementioned competitive learning protocols. Thus, as can be seen in Table **??**, the competitors' performances are gradually enhanced as the number of participating generators increases. U-IncNet (vs. ResNet) achieves the best performance in terms of PSNR/SSIM, with top scores of 31.73/0.9624, followed by U-IncNet (vs. U-Net) with scores of 31.67/0.9598. The proposed competitive adversarial learning approach enhances the performance of all competing generators. When competing individually against U-Net and U-IncNet generators, the PSNR/SSIM scores of ResNet are improved by 2.75%/0.51% and 4.98%/1.32%, respectively. Furthermore, when competing against U-Net and U-IncNet simultaneously, ResNet's performance is improved by 7.45% and 2.28% in terms of PSNR and SSIM, respectively. On the other hand, U-Net in competition against ReNet and U-IncNet simultaneously yields the best PSNR/SSIM scores of 31.41/0.9584 compared to its scores following other learning scenarios. As a whole, the results achieved for image Demoireing verifies the fact that competitive adversarial learning, in which all generators compete to provide their best results, improves the generation performance.

4.1.4 visible to Infrared image translation

Due to the complexity associated with capturing Infrared images, several deep learning approaches have been developed to convert visible images to Infrared images synthetically. The selected benchmark dataset of such task is LLVIP dataset [34] which consists of 15488 pairs of images (visible and infrared) which were taken under different low-lighting conditions. Similarly, the state-of-the-art deep learning approaches proposed for this task are evaluated based on their PSNR and SSIM values.

Table **??** lists the PSNR/SSIM scores estimated on LLVIP dataset. The single learning based configurations achieve low scores compared to the scores achieved following the competitive adversarial learning protocols. For example, ResNet achieves PSNR/SSIM scores as low as 17.08/0.4595 when trained individually whereas these scores improve by 15.86%/35.55% (respectively) when trained competitively against other generators. Also, U-Net boosts its PSNR score from 22.21 to 22.95 (gain of 3.37%) following *one-vs-all* learning protocol compared to a slight enhancement (gain of 0.07%) achieved following one-vs-one learning scenario (U-net vs U-IncNet). Likewise, U-IncNet's performance is significantly improved and it scores the highest PSNR/SSIM values of 23.21/0.7662 following a *one-vs-all* learning protocol with gains of 8.35%/10.14% in terms of PSNR and SSIM, respectively. We also notice that U-Net's performance is not robust against ResNet, in contrary to U-IncNet that improves its performance even when trained in competitively against ResNet thanks to the extra feature extraction block. Due to the challenges associated with infrared translation application, it can be inferred from Table **??** that the reported PSNR/SSIM are low compared to previous experiments.

4.1.5 Deblurring

Image deblurring is the process where sharp images with fine details are recovered from motion blurred degraded images. Such degradation is mainly caused by the camera motion occurred while capturing an image using handful devices such as mobile phones and mounted cameras. Single image deblurring has become a hot topic in the field of deep learning where it is still addressed by several deep learning approaches. The selected benchmark dataset in this experiment is Human-aware Image deblurring HIDE which was introduced in [35]. HIDE dataset contains 8,482 pairs of motion-blurred images next to their corresponding sharp images, where both are extracted from outdoor videos captured at 31 frames per second. The images are degraded by wide range of motion patterns and are split into training subset (6,397 images) and testing subset (2,025 images). Table **??** shows the quantitative results obtained by our baseline competitors on HIDE blur dataset. U-IncNet-vs-all achieves the highest PSNR/SSIM scores of 31.98/0.9587 with a gain of 11.47%/6.31%, respectively. From Table **??**, it can be seen that all competing learning protocols lead to better results than standalone configurations, which proves the effectiveness and the superiority of the proposed CMGANs. ResNet baseline proved to be more motivated for improving its performance when trained in competition against the other generators. Its performance is enhanced by 17.61% in terms of PSNR (from 25.61 to 30.12) when competing against U-IncNet, and by a gain of 7.01% in terms of SSIM (from 0.866 to 0.927) when using the *one-vs-all* competitive learning configuration. Similarly, U-Net increases its PSNR/SSIM scores by 13.65%/6.40% and 11.60%/4.60% following the *one-vs-all* and *one-vs-ResNet* learning strategies, respectively. The *U-Net-vs-U-IncNet* configuration scores 28.33/0.8997, leading to an improvement of 7.10%/3.31% compared to U-Net's single score.

4.1.6 Dewatering

Due to bad capturing conditions such as light condition, type of water, and depth, the images captured underwater suffer from severe degradation in color tubes. To handle such issue, several deep learning approaches have been proposed to enhance underwater images where the majority of them are based on different GANs variants. Such approaches including CMGANs are trained using a dataset of 6128 paired images where the GT images are synthesized using CycleGANs [36]. On the other hand, the evaluated

approaches are tested on image Enhancement Benchmark Dataset (UIEBD) [37] that contains 890 underwater images that suffer from low contrast, color shift and haze based degradation. Due to the absence of realistic ground-truths, the proposed deep learning appraoches are evaluated using non-reference evaluation metrics such as underwater color image quality evaluation (UCIQE) [38] and underwater image quality measure (UIQM) [39] where the highest the values, the better the results are. The scores of the dewatering evaluation metrics UCIQE/UIQM are listed in Table **??** where the best result is achieved by U-IncNet following *one-vs-all* learning scenario with 0.8831 UCIQE and 6.6105 UIQM improving its standalone performance by 35.90% and 26.28%, respectively. Following the competitive learning strategy, ResNet also improves its learning performance where its UCIQE gain ranges from 46.22% to 59.97% and from 24.45% to 46.79% with respect to UIQM metric. Moreover, U-Net's performance is the most improved one by the proposed competitive learning where it reaches 64.31% of gain on UCIQE score and 54.34% of gain on UQIM score. This experiment proves that the proposed competitive learning protocol leads to important improvements compared to the single-based learning one especially when applied on traditional and multi-purpose generators like U-Net and ResNet. Taking into consideration its good performance through the single mode, the proposed U-IncNet experiences less improvements compared to the other competitors trained following the competitive learning strategy.

4.2 Discussion and remarks

Like any competition based experiment, it is expected that the competing generators will show different degrees of motivation and varying performances when following different competitive adversarial scenarios. In other words, some generators could be positively motivated by competition against other generators where their performances will be improved compared to their stand-alone ones. On the other hand, some generators are expected to be negatively affected by competition so their performances could slightly be degraded compared to their stand-alone performances. We name the former generators as task-oriented generators and the latter ones as ego-oriented generators. Based on the quantitative results reported above, U-Net generator can be identified as an ego-oriented generator where its performance is slightly downgraded following the competitive learning scenarios as can be noted in deraindroping, Demoireing, and Infrared translation experiments. On the other hand, generators such as ResNet and U-IncNet show a task-oriented competition behaviour where their performances are enhanced following the competitive learning protocols in most of the targeted applications. It is worth mentioning that among the tested generators, ResNet is greatly motivated when competing against all other generators where for example, its performance shows large gain rates of PSNR compared to other generators (7.45%, in demoireing, 15.86% in infrared translation, and 17.06% in deblurring).

In experiments such as infrared image translation, deblurring, and dewatering; the competing generators show better performances when they are trained following *one-vs-all* learning protocol compared to their performances achieved by following *one-vs-one* scenario. While, the remaining experiment show no significant improvement in the competitors' performances when following different competitive adversarial learning protocols. These facts indicate that in certain applications increasing the number of participating competitors significantly improves their performances and learning processes. This can be observed -for example- in Fig. **??** (a) and (b) that demonstrates how effective the competitive learning is in minimizing the loss faster than the single learning protocol on the HIDE blur and LLVIP benchmarks. Several significant and interesting observations can be made out of Fig. **??** such as the different learning behavior that each competitor shows under different competitive learning strategies. For example, it can be seen in both experiments that ResNet's learning phase following *one-vs-all* scenario; starts with high loss value due to the motivation penalties added to its loss to outperform other generators. Such penalties motivates ResNet to provide better results that minimize the learning loss with significant margin compared to the one achieved during standalone and *one-vs-one* learning scenario. On the other hand, U-Net's learning behavior does not show such severe drop in the loss as it is rarely penalized by the loss differences due to its ego-oriented character and its good performance while competing. However, its learning curve shows significant improvement in the loss minimization ability when trained following both competitive learning strategies in the deblurring experiment while it shows no improvement in the to-infrared image translation experiment when trained competitively. In contrary, U-IncNet shows almost similar learning behaviors in the deblurring experiment regardless the followed learning strategies Thanks to its effective feature extraction phase. However, in the to-infrared image translation experiment where there is still room for improvement, U-IncNet following both competitive learning strategies achieves significant improvement compared to the stand-alone scenario.

4.3 Qualitative assessment

The qualitative evaluation of the three generators helps to better understand visually the effect of the proposed competitive learning on the estimated results. In Figure **??**, two visual examples (with noticeable improvements corresponding to the gains discussed in the ablation study) of each generator are shown. From the results estimated on LLVIP dataset ((see Fig. **??** (1st row)), we can clearly see the significant visual improvements achieved by ResNet when trained competitively compared to the results estimated on stand-alone basis where the images are noisy and pedestrians/cars are mistranslated into infrared domain. The ResNet's best visual result is achieved via one-vs-all protocol. The same ResNet's behavior is also noticed on Dewater dataset, where the less blue-tone images with correct illumination/contrast are obtained following the *one-vs-all* learning protocol (see Fig. **??** (4th row)). Likewise, images generated following ResNet vs. U-Net scenario are less blue-toned but brighter than the target one. With respect to U-Net generator, the most visual improvements are found when tested on the TIP2018 and HIDE benchmarks. Despite the severe moiré patterns and low contrast presented in the

input image, the U-Net vs. All scenario manages to correct the contrast and significantly removes the patterns while maintaining good image quality (see Fig. **??** (2nd row)). The images corrected by other U-Net's configurations are well demoiréized but suffer from quality drop with pixelated regions around the projector. The significant improvement in U-Net's deblurring ability is achieved with U-Net vs. U-IncNet configuration as it recovers images with more face details compared to the other configurations (see Fig. **??** (5th row)). For visual Desnowing results, the baseline models and their competitive configurations are applied on realistic images without groundtruths. The image obtained by U-IncNet trained individually contains more snowflakes compared to the ones obtained following *one-vs-one* scenarios while the best visual result is produced by U-IncNet vs. All configuration (see Fig. **??** (3rd row)). With respect to the deraindroping experiment where the generators have to correctly fill the gaps caused by the raindrop removal, U-IncNet vs. All configuration manages to successfully remove (see **??** (6th row)) the drops especially on the right building and the sky. The second best result is achieved by U-IncNet vs. U-Net configuration, while the alone mode removes most of the drops but couldn't deal with their depth effects. Based on all above, we can disclose that the proposed MCGANs framework allows the baseline generators to deliver very enhanced images on six challenging applications.

4.4 Comparison against the state-of-the-art (SOTA)

In this section, we present an overall evaluation of the performance of our proposed MCGANs compared to the current state-of-the-art approaches. Tables **??** and **??** show the performance comparisons with well-known existing approaches in the targeted applications. From both tables, it is observed that the proposed competitive learning approach allows the three baseline generators to achieve higher metrics compared to the state-of-the-art techniques on the majority of the benchmarks. For image demoireing, it is clear that U-IncNet following the three competitive learning strategies outperforms all state-of-the-art approaches. U-IncNet (vs. ResNet) restores perfectly clean versions of images affected by moiré patterns with a high PSNR/SSIM scores of 31.89/0.9624, outperforming the best SOTA approach MBCNN by 1.86dB/7% in terms of PSNR and SSIM respectively. In deraindroping experiment, it is noted that most approaches -more or less- show competitive performances with respect to PSNR scores. The experimental results of ResNet based MCGANs are superior to all those reported in the literature. ResNet-vs. all configuration achieves the highest PSNR/SSIM scores of 31.85/0.9562, surpassing the best SOTA approach AGAN by 5.39% in terms of SSIM score and a gain of 0.28 dB in terms of PSNR score. In image desnowing, U-IncNet-vs. U-Net records the highest PSNR/SSIM of 31.53/0.945, outperforming all state-of-the-art approaches. These results confirm the findings from Figure **??**, where the proposed competitive adversarial learning generated snow-free images with less blur and more fine details, leading to the most visually appealing results. In deblurring experiment, U-IncNet-vs. all outperforms the state-of-the-art techniques with top PSNR/SSIM scores of 31.98/0.958, which exceeds the scores achieved by the best SOTA method by +0.56dB/1.4%. In the infrared image translation experiment, U-IncNet-vs. all, U-Net (vs. All), and U-IncNet (vs. ResNet) configurations clearly outperform the Pix2Pix based GANs and provide a reliable solution for synthesising visible images into infrared images with high accuracy. They deliver significant performance improvements of +12.45dB/59.05%, +12.19dB/56.16%, and +11.65/57.91% over the Pix2Pix, respectively. Finally, in the dewatering experiment, all competing generators when following the *one-vs-all* learning protocol, achieve excellent results in terms of UCIQE and UIQM socres, ranking first ahead of the the state-of-the-art techniques. As can be noted in Table **??**, our U-IncNet (vs. all) outperforms DewaterNet, the best SOTA technique, by a significant UCIQE/UQIM margins. This confirms the results in Figure **??**, where our proposal shows accurate comprehensive performance in the illustrated underwater scene, especially in removing the blue effect.

4.5 Implementation details

The proposed MCGANs framework is developed using the Python3.9 along with Pytorch 1.7 Neural Network Libraries and CUDA GPU Toolkit 11, and performed on Alienware Aurora R11 i9-10900KF Dual RTX2080Ti (22GB VRAM).

5 Conclusion

In this paper, we proposed a new competition based adversarial learning in which the protocol rewards or penalizes every competitor based on its its performance (win or loss) against other competitors. This competitive environment provides more motivation for the competitors to learn more and faster. We adopted the Pix2Pix architecture for both U-Net and ResNet based generators in addition to an introduced -in this paper- variant of U-Net enhanced with a feature extraction block derived from Inception.v3 model (U-IncNet). We considered the pixel-wise discriminator as a referee over the three generators. Moreover, L1 loss distance is employed to judge the participating generators' individual performance so rewards or penalties can be assigned to the competitors. The comprehensive ablation study performed on six challenging applications covering image enhancement, infrared image translation, and bad-weather removal, proved that the proposed MCGANs framework helps the baseline generators to significantly improve their performance compared to the stand-alone learning approach. The obtained performance's gains are very high on most of the benchmarks reaching 64.31% as in the case of the UCIQE dewatering metric and 35.99% over the SSIM as in the case of infrared translation application. Moreover, the proposed MCGANs framework manages to improve baseline architectures such as ResNet and U-Net so they outperformed several state-of-the-art techniques. This fact proves that our proposed learning protocol can be very beneficial for the community to improve performances of previously proposed approaches on other paired image generation tasks especially if new competitors and referees are developed. As instance, the proposed U-IncNet helps other generators to achieve better results. Therefore, developing more competitive generators would be an interesting

opening as a future work. In addition, we intend to define application-wise referees through learning-based loss functions and discriminators instead of the L1 and traditional losses.

References

[1] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in neural information processing systems*, vol. 27, 2014.

[2] S. Wen, W. Liu, Y. Yang, T. Huang, and Z. Zeng, "Generating realistic videos from keyframes with concatenated gans," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 29, no. 8, p. 2337–2348, Aug 2019.

[3] D. Chen, L. Yue, X. Chang, M. Xu, and T. Jia, "Nm-gan: Noise-modulated generative adversarial network for video anomaly detection," *Pattern recognition*, vol. 116, p. 107969, Aug 2021.

[4] D.-W. Jaw, S.-C. Huang, and S.-Y. Kuo, "Desnowgan: An efficient single image snow removal framework using cross-resolution lateral connection and gans," *IEEE Transactions on Circuits and Systems for Video Technology*, p. 1–1, 2020.

[5] T.-C. Wang, M.-Y. Liu, J.-Y. Zhu, A. Tao, J. Kautz, and B. Catanzaro, "High-resolution image synthesis and semantic manipulation with conditional gans," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 8798–8807.

[6] H. Emami, M. M. Aliabadi, M. Dong, and R. B. Chinnam, "Spagan: Spatial attention gan for image-to-image translation," *IEEE Transactions on Multimedia*, vol. 23, pp. 391–401, 2020.

[7] Z. Yi, H. Zhang, P. Tan, and M. Gong, "Dualgan: Unsupervised dual learning for image-to-image translation," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2849–2857.

[8] W. Lira, J. Merz, D. Ritchie, D. Cohen-Or, and H. Zhang, "Ganhopper: Multi-hop gan for unsupervised image-to-image translation," in *European Conference on Computer Vision*. Springer, 2020, pp. 363–379.

[9] H. Zhang, V. Sindagi, and V. M. Patel, "Image de-raining using a conditional generative adversarial network," *IEEE transactions on circuits and systems for video technology*, vol. 30, no. 11, pp. 3943–3956, 2019.

[10] H. Phan, I. V. McLoughlin, L. Pham, O. Y. Chén, P. Koch, M. De Vos, and A. Mertins, "Improving gans for speech enhancement," *IEEE Signal Processing Letters*, vol. 27, pp. 1700–1704, 2020.

[11] A. Ghosh, V. Kulharia, and V. Namboodiri, "Message passing multi-agent gans," *arXiv preprint arXiv:1612.01294*, 2016.

[12] Q. Hoang, T. D. Nguyen, T. Le, and D. Phung, "Mgan: Training generative adversarial nets with multiple generators," in *International conference on learning representations*, 2018.

[13] T. Chavdarova and F. Fleuret, "Sgan: An alternative training of generative adversarial networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 9407–9415.

[14] A. Ghosh, V. Kulharia, V. P. Namboodiri, P. H. Torr, and P. K. Dokania, "Multi-agent diverse generative adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 8513–8521.

[15] I. Durugkar, I. Gemp, and S. Mahadevan, "Generative multi-adversarial networks," *arXiv preprint arXiv:1611.01673*, 2016.

[16] C. Hardy, E. Le Merrer, and B. Sericola, "Md-gan: Multi-discriminator generative adversarial networks for distributed datasets," in *2019 IEEE international parallel and distributed processing symposium (IPDPS)*. IEEE, 2019, pp. 866–877.

[17] S. Arora, R. Ge, Y. Liang, T. Ma, and Y. Zhang, "Generalization and equilibrium in generative adversarial nets (gans)," in *International Conference on Machine Learning*. PMLR, 2017, pp. 224–232.

[18] M. Rasouli, T. Sun, and R. Rajagopal, "Fedgan: Federated generative adversarial networks for distributed data," *arXiv preprint arXiv:2006.07228*, 2020.

[19] H. Zhang, S. Xu, J. Jiao, P. Xie, R. Salakhutdinov, and E. P. Xing, "Stackelberg gan: Towards provable minimax equilibrium via multi-generator architectures," *arXiv preprint arXiv:1811.08010*, 2018.

[20] W. Li, L. Fan, Z. Wang, C. Ma, and X. Cui, "Tackling mode collapse in multi-generator gans with orthogonal vectors," *Pattern Recognition*, vol. 110, p. 107646, 2021.

[21] Y. Wang, A. Gonzalez-Garcia, D. Berga, L. Herranz, F. S. Khan, and J. v. d. Weijer, "Minegan: effective knowledge transfer from gans to target domains with few images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 9332–9341.

[22] C. Wang, C. Xu, X. Yao, and D. Tao, "Evolutionary generative adversarial networks," *IEEE Transactions on Evolutionary Computation*, vol. 23, no. 6, pp. 921–934, 2019.

[23] T. M. Quan, T. Nguyen-Duc, and W.-K. Jeong, "Compressed sensing mri reconstruction using a generative adversarial network with a cyclic loss," *IEEE transactions on medical imaging*, vol. 37, no. 6, pp. 1488–1497, 2018.

[24] H. Tang, X. Qi, D. Xu, P. H. Torr, and N. Sebe, "Edge guided gans with semantic preserving for semantic image synthesis," *arXiv preprint arXiv:2003.13898*, 2020.

[25] S. Wen, W. Liu, Y. Yang, T. Huang, and Z. Zeng, "Generating realistic videos from keyframes with concatenated gans," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 29, no. 8, pp. 2337–2348, 2018.

[26] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[28] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 1–9.

[29] R. E. Franken and D. J. Brown, "Why do people like competition? the motivation for winning, putting forth effort, improving one's performance, performing well, being instrumental, and expressing forceful/aggressive behavior," *Personality and individual differences*, vol. 19, no. 2, pp. 175–184, 1995.

[30] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1125–1134.

[31] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, and A. A. Efros, "Context encoders: Feature learning by inpainting," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2536–2544.

[32] R. Qian, R. T. Tan, W. Yang, J. Su, and J. Liu, "Attentive generative adversarial network for raindrop removal from a single image," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 2482–2491.

[33] Y. Sun, Y. Yu, and W. Wang, "Moiré photo restoration using multiresolution convolutional neural networks," *IEEE Transactions on Image Processing*, vol. 27, no. 8, pp. 4160–4172, 2018.

[34] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "Llvip: A visible-infrared paired dataset for low-light vision," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 3496–3504.

[35] Z. Shen, W. Wang, X. Lu, J. Shen, H. Ling, T. Xu, and L. Shao, "Human-aware motion deblurring," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 5572–5581.

[36] C. Fabbri, M. J. Islam, and J. Sattar, "Enhancing underwater imagery using generative adversarial networks," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2018, pp. 7159–7165.

[37] C. Li, C. Guo, W. Ren, R. Cong, J. Hou, S. Kwong, and D. Tao, "An underwater image enhancement benchmark dataset and beyond," *IEEE Transactions on Image Processing*, vol. 29, pp. 4376–4389, 2019.

[38] M. Yang and A. Sowmya, "An underwater color image quality evaluation metric," *IEEE Transactions on Image Processing*, vol. 24, no. 12, pp. 6062–6071, 2015.

[39] K. Panetta, C. Gao, and S. Agaian, "Human-visual-system-inspired underwater image quality measures," *IEEE Journal of Oceanic Engineering*, vol. 41, no. 3, pp. 541–551, 2015.